

SOC L1 Readiness Test (Interview Preparation)

Includes Questions, Answers, and Explanations. Designed for SOC L1 training and interview preparation.

Section A — MCQ (Fundamental Knowledge)

1. What is the primary role of a SOC L1 analyst?

Answer: B. Monitor alerts and perform initial triage

Explanation: SOC L1 analysts monitor security alerts, validate them, perform initial investigation, and escalate incidents.

2. Which malware encrypts files and demands payment?

Answer: C. Ransomware

Explanation: Ransomware encrypts victim files and demands payment for decryption.

3. Which attack uses one password across many accounts?

Answer: C. Password Spraying

Explanation: Password spraying tries one common password across many users to avoid lockouts.

4. What does hashing primarily provide?

Answer: B. Integrity

Explanation: Hashing ensures integrity by producing a unique fingerprint of data.

5. What does salting prevent?

Answer: B. Rainbow table attacks

Explanation: Salting randomizes hashes so attackers cannot rely on precomputed tables.

6. Which tool is commonly used for credential dumping?

Answer: C. Mimikatz

Explanation: Mimikatz extracts credentials and hashes from Windows memory.

7. Which Windows process stores credentials in memory?

Answer: C. lsass.exe

Explanation: LSASS handles authentication and stores credential material.

8. Which attack redirects users to a fake website without clicking malicious links?

Answer: B. Pharming

Explanation: Pharming manipulates DNS to redirect traffic to malicious sites.

9. Moving from one compromised machine to another inside a network is called:

Answer: C. Lateral Movement

Explanation: Attackers pivot internally to reach high-value systems.

10. Which encryption type uses two keys?

Answer: B. Asymmetric encryption

Explanation: Public and private keys are used.

11. Which attack involves sending fraudulent emails to steal credentials?

Answer: A. Phishing

Explanation: Phishing uses deceptive communication to trick victims.

12. Which technique steals password hashes from memory?

Answer: A. Credential Dumping

Explanation: Attackers extract credentials stored in system memory.

13. Which event ID represents failed login attempts in Windows?

Answer: A. 4625

Explanation: Event ID 4625 indicates failed logon attempts.

14. What type of malware secretly collects user information?

Answer: B. Spyware

Explanation: Spyware monitors activity and steals sensitive information.

15. Which attack targets specific individuals using personalized information?

Answer: B. Spear Phishing

Explanation: Attackers tailor messages using reconnaissance.

16. What does IOC stand for?

Answer: B. Indicator of Compromise

Explanation: Artifacts indicating potential malicious activity.

17. Which attack involves creating a fake WiFi network to capture data?

Answer: C. Evil Twin

Explanation: Attackers mimic legitimate WiFi networks.

18. What is the main purpose of SIEM systems?

Answer: B. Log collection and security monitoring

Explanation: SIEM aggregates and analyzes security events.

19. Which attack floods a system with traffic to make it unavailable?

Answer: A. DDoS

Explanation: Distributed Denial of Service overwhelms resources.

20. Which authentication method uses multiple verification factors?

Answer: C. Multi-Factor Authentication

Explanation: Combines multiple verification mechanisms.

21. Which process involves analyzing alerts to determine severity?

Answer: B. Alert Triage

Explanation: SOC analysts evaluate alerts for threat relevance.

22. What type of privilege escalation occurs when a user becomes administrator?

Answer: A. Vertical Privilege Escalation

Explanation: Privilege increases from low to high level.

23. Which attack reuses stolen username/password pairs from breaches?

Answer: B. Credential Stuffing

Explanation: Automated login attempts using leaked credentials.

24. What security concept ensures data has not been altered?

Answer: C. Integrity

Explanation: Integrity guarantees unchanged data.

25. Which protocol is commonly targeted for remote desktop access?

Answer: B. RDP

Explanation: Remote Desktop Protocol enables remote Windows login.

Section B — Short Answer Questions

Explain the difference between hashing and encryption.

Answer: Hashing is one-way and mainly used for password storage and integrity verification. Encryption is reversible using keys and is used to protect confidential data.

What is password spraying and why is it difficult to detect?

Answer: Password spraying uses one common password across many accounts. Because it tries only one password per account, lockouts are avoided.

What is lateral movement and why do attackers perform it?

Answer: Lateral movement occurs when attackers move across internal systems after initial compromise to reach high-value assets.

Explain vertical vs horizontal privilege escalation.

Answer: Vertical escalation increases privileges (user → admin). Horizontal escalation accesses another account with similar privilege.

What is credential stuffing?

Answer: Credential stuffing uses previously leaked username-password combinations to access other services.

Explain the concept of salting in password storage.

Answer: Salting adds random data before hashing passwords to prevent rainbow table attacks.

What are Indicators of Compromise (IOC)? Give examples.

Answer: IOCs are artifacts indicating malicious activity such as malicious IPs, file hashes, or suspicious domains.

Explain the difference between phishing and spear phishing.

Answer: Phishing targets many users, spear phishing targets specific individuals.

What is an Evil Twin attack?

Answer: Attackers create a fake WiFi network mimicking a legitimate one to intercept traffic.

What is alert triage in SOC operations?

Answer: Alert triage is the process of reviewing and prioritizing alerts to determine whether they represent real incidents.

Section C — Scenario Based Questions

Logs show 500 failed login attempts from the same IP targeting one account.

Answer & Explanation: Brute Force Attack. Many password attempts against one account.

One password attempted across 200 user accounts.

Answer & Explanation: Password Spraying attack.

Mimikatz executed, LSASS accessed, admin login occurs, RDP to file server.

Answer & Explanation: Credential Dumping → Privilege Escalation → Lateral Movement.

Employee receives email: 'Your account will be suspended. Click here to reset password.'

Answer & Explanation: Phishing attempt.

Login from India followed by login from Russia within 3 minutes.

Answer & Explanation: Impossible travel indicator suggesting possible account compromise.

Section D — Analyst Thinking

What steps would you take when investigating a suspicious login alert?

Answer: Check IP address, geolocation, user behavior, device info, login time, and authentication logs.

If a user clicked a phishing email, what should the SOC analyst do first?

Answer: Reset credentials, isolate the system if necessary, scan endpoint, and review logs.

Why do attackers try to escalate privileges after gaining initial access?

Answer: Higher privileges allow attackers to disable security controls and access sensitive systems.

How would you differentiate between a false positive and a real incident?

Answer: Correlate logs, check context, analyze user behavior, and validate activity.

What logs would you check when investigating a brute force attack?

Answer: Windows security logs, authentication logs, VPN logs, firewall logs, and SIEM alerts.